

Back propagation:- (B.P through time):-

We will take Many to 1 as eg
on sentiment analysis.

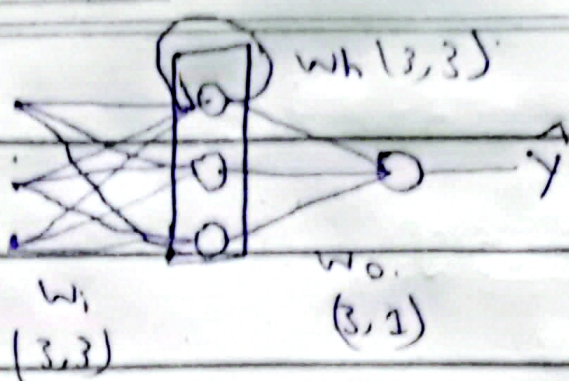
text.

				x	y
Cat	mat	rat	1	x_1 [1 0 0] [0 1 0] [0 0 1]	1
rat	rat	met	1	x_2 [0 0 1] [0 0 1] [0 1 0]	1
met	met	Cat	0	x_3 [0 1 0] [0 1 0] [1 0 0]	0

Vocab $\rightarrow$ Cat Met Rat
[1 0 0] [0 1 0] [0 0 1]

Date: _____

Day: _____



→ Send these weights to nodes & calculate loss.

→ To find loss we need to minimize the loss through → Gradient descent.

which is actually...

$$w_i = w_i - \eta \frac{\partial L}{\partial w_i}$$

$$w_h = w_h - \eta \frac{\partial L}{\partial w_h}$$

$$w_o = w_o - \eta \frac{\partial L}{\partial w_o}$$

So we will find these derivative along with we decide learning rate and old weight value, so updated value with min loss comes.

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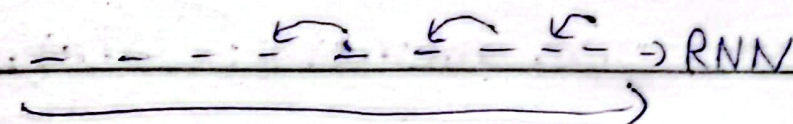
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Problems with RNN:-

RNNs $\rightarrow$ Seq. data $\rightarrow$ textual, time-series

$\downarrow$
suffer 2 major problems $\rightarrow$ LSTMs

- $\rightarrow$ Problem of long term dependence.
 - $\rightarrow$ Unstable gradients / Training.
- } Major problems.



1st prob:-

- $\rightarrow$ If a particular timestamp thing depends on very old timestamp thing $\rightarrow$ RNN haven't that capability of long term $\rightarrow$ RNN fails $\rightarrow$ Vanishing grad problem.

If we have 100 timestamp.

$$\frac{\partial L}{\partial w_{in}} = \frac{\partial L}{\partial \vec{y}} \cdot \frac{\partial \vec{y}}{\partial o_{100}} \cdot \frac{\partial o_{100}}{\partial o_{99}} \cdots \frac{\partial o_2}{\partial i_1} \cdot \frac{\partial o_1}{\partial w_{in}}$$

and that's forgetting about past or using it less is van. gradient problem.

Date: _____

Day: _____

Possible solutions:-

- 1) Diff activation $\rightarrow$ relu/leaky relu.
- 2) Better weight initialization.
- 3) Skip rnn's. 4) LSTM.

So leave RNN & let's move to
LSTM.